PingPong: A Benchmark for Role-Playing Language Models with User Emulation and Multi-Model Evaluation

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Abstract

We introduce a benchmark for evaluating the role-playing capabilities of language models. Our approach leverages different language models to simulate users in dynamic, multi-turn conversations and assess the resulting dialogues. Our methodology involves three main components: a player model that adopts a specific character role, an interrogator model that simulates user behavior in a specific situation, and a judge model ensemble that evaluates conversation quality with 3 metrics: character consistency, entertainment value, and language fluency. We evaluated more than 40 models in both English and Russian, with each model participating in 64 conversations with 8 characters and 8 situations. We conducted experiments comparing automated evaluations with human annotations to validate our approach, demonstrating strong correlations across multiple criteria. This work provides a foundation for a robust and dynamic evaluation of different model capabilities in interactive scenarios.

1 Introduction

Language models, which predict plausible language, have dominated natural language processing since BERT (Devlin et al., 2019), with models like ChatGPT (Ouyang et al., 2022) showcasing advanced conversational capabilities.

In this paper, we focus on role-playing language models for entertainment purposes. These models are assigned specific characters or personas and are tasked with maintaining these roles while engaging and entertaining users. Although there are other important applications of role-playing language models, such as training mental health specialists (Louie et al., 2024) or simulating human opinion dynamics (Chuang et al., 2024), they are beyond the scope of this paper.

People seek entertainment through interactions with their favorite characters from games, movies, and TV shows. The quality of these interactions directly impacts user satisfaction and engagement. As companies continue to develop dedicated platforms for role-playing language models, such as Character.ai[1] or Chai(Irvine et al., 2023), robust evaluation frameworks are critical. Developers of such systems cannot effectively improve them or compare competing approaches without standardized methods to assess how well models maintain character consistency and entertain users.

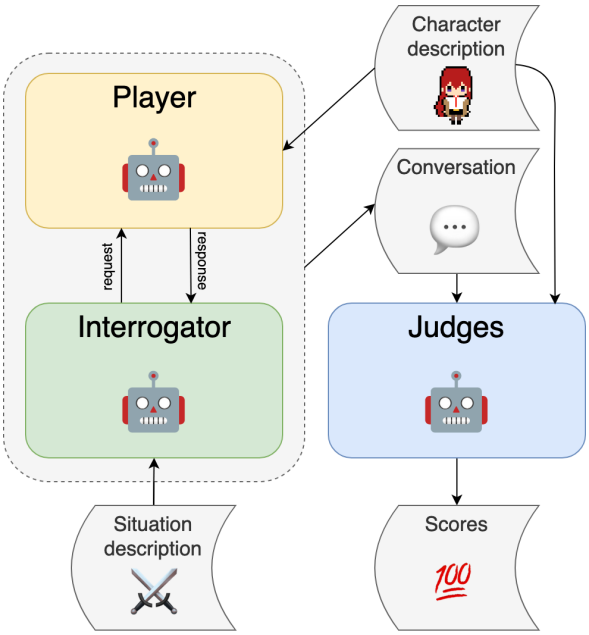

Figure 1: This diagram illustrates the flow of interactions in the proposed benchmark. There are three main components with different language models: a player, an interrogator, and a judge ensemble. The player assumes some character role, the interrogator acts as a user in a specific situation, and the judges evaluate final conversations.

To fill this gap, we introduce a novel benchmark for evaluating role-playing language models. From our experience with language models, we believe that direct interaction is the most effective way to assess a language model's conversational abilities.

[1] https://character.ai

However, humans often lack time to test new models manually, and many popular benchmarks are limited to single-turn interactions (Dubois et al., 2024; Hendrycks et al., 2021). These benchmarks are also becoming less reliable due to test data contamination (Deng et al., 2024). To address this, we propose using language models to simulate users in role-playing conversations and automatically evaluate the resulting dialogues.

Our methodology, illustrated in Figure 1, involves three key components: a player model that assumes a character role, an interrogator model that simulates user behavior, and a judge model that evaluates the quality of the conversation. Our work builds on existing benchmarks, such as the EQ-bench (Paech, 2023), introducing an approach to evaluate the role-playing abilities of language models.

Our contributions:

- We propose a benchmark for assessing the role-playing abilities of language models. The **combination** of the following traits makes it novel:
  - **Multi-turn**: All conversations have multiple turns to be closer to the real usage of role-playing models.
  - **Dynamic**: The interrogator questions are generated by sampling language models and are not predefined. Each evaluation run produces different questions, making it harder for models to memorize responses and making test data contamination harder.
  - **Multi-model**: Several judges mitigate individual model biases and correlate better with humans. It also allows for the evaluation of the models that are used as judges.
- We validate our benchmark through correlation with manual annotations and comparison with other benchmarks.
- We show that the multi-model setup correlates better with humans than a single model.
- We discover that fine-tuning models for creative writing improves their role-playing abilities.

All results, prompts, and scripts are available online[2].

The benchmark website has up-to-date leaderboards[3] and all conversations with example-wise scores, which allows the eyeballing of common errors and problems of evaluated models. It is available for English and Russian languages.

2 Related work

Role-playing capabilities and evaluation. Various commercial services exploit the role-playing abilities of language models, including Character.ai and Chai. Academic and community efforts are underway to create similar systems with open datasets, code, and models, such as PIPPA, ChatHaruhi, Character-LLM (Gosling et al., 2023; Li et al., 2023; Shao et al., 2023), MythoMax[4], or Magnum[5]. Several static benchmarks for role play exist, including ECHO, InCharacter, RAIDEN, and CharacterEval (Ng et al., 2024; Wang et al., 2024; Wu et al., 2025; Tu et al., 2024).

PersonaGym (Samuel et al., 2024) is close to our work, featuring dynamic question generation based on the environment ("situation" in our terminology) and the currently selected persona. There is also a very similar dynamic benchmark, RPBench-Auto[6]. It is based on the same assumptions and features and has a structure similar to one of the versions of our benchmark. The significant difference from our work is that their evaluation is based on side-by-side comparisons with the baseline model responses, while we produce single-point evaluations. One can also utilize evaluation based on multiple choice questions similar to the one developed in the DITTO framework (Lu et al., 2024). Although it efficiently assesses the role consistency, the evaluation is based heavily on questions based on artificial knowledge that may not reflect typical user interactions with role-playing systems.

One common problem for most of these benchmarks is a lack of long-term support and reproducibility, which might be caused by the high cost of evaluations. Regularly adding new models to a benchmark is crucial, as this is the only way the benchmark can guide practical decisions.

A different approach to evaluation would be to use online metrics, such as retention rates or user ratings (Irvine et al., 2023). However, this approach is only viable if you already have a service with a

[2] https://github.com/ilyagusev/ping_pong_bench

[3] https://ilyagusev.github.io/ping_pong_bench

[4] https://huggingface.co/Gryphe/MythoMax-L2-13b

[5] https://huggingface.co/anthracite-org/magnum-v2-123b

[6] https://boson.ai/rpbench-blog

substantial user base.

Automatic evaluation with language models. LLM-as-a-Judge (Zheng et al., 2023) is an evaluation method that relies on language models, such as GPT-4, instead of humans. Popular benchmarks using this method include AlpacaEval, EQ-bench, Creative Writing, and BiGGen Bench (Dubois et al., 2024; Paech, 2023; Kim et al., 2024). The validity of these benchmarks is based on their high correlation with human annotations, specifically with Chatbot Arena (Chiang et al., 2024).

Multi-model evaluation. However, all these benchmarks are based on a single model as a judge, which may introduce various biases, including self-evaluation bias (Panickssery et al., 2024; Xu et al., 2024). PoLL (Verga et al., 2024) authors aggregate evaluations from different language models similarly to us, with average pooling. They show that ensembling different models for evaluation increases the correlation with human annotations. There is another more agentic approach (Chan et al., 2024) with a referee team.

Multi-turn evaluation. Most benchmarks are single-turn, which contrasts with the real-world usage of language models. There are multi-turn benchmarks, such as MT-Bench-101 (Bai et al., 2024) and MT-Eval (Kwan et al., 2024), though they focus on specific capabilities, and their evaluation procedures still differ from how humans implicitly rate language models.

Data contamination. Another major problem for static public benchmarks is data leakage into the pre-training data sets of language models (Deng et al., 2024). It is challenging to avoid contamination since such tests are usually stored online and considered "code" during pre-training. This can occur even with no malicious intent from the model creators. The most obvious solution is to close the benchmarks completely, which requires trusting the benchmark organizers, which is difficult in a highly competitive environment. Alternative solutions include regularly updating benchmarks with new test data (White et al., 2024) or dynamically generating test data using existing language models.

3 Methodology

3.1 Role definitions

Our framework comprises three principal roles: player, interrogator, and judge, inspired by the Turing test (Turing, 1950). However, our approach differs in the number of agents, the player's objective, and the use of machine-based interrogators and judges.

Language models can assume three possible roles.

- **Player** assumes the role of a specific character based on a character card provided.
- **Interrogator** engages the player within a given situation or towards a specific goal, simulating user behavior.
- **Judge** evaluates the player's responses against predetermined criteria.

Role assignments are implemented through a combination of system and user prompts. We use only models that support chat templates. All instructions are incorporated into the user prompt for models without dedicated system prompts, such as Gemma 2 (GemmaTeam, 2024).

This setup is **asymmetric** since the player only gets the character description, while the interrogator only gets the situation information. This is intentional, as typical use cases of role-playing models are asymmetrical. However, it is possible to modify it to make it symmetric by providing character descriptions and situations to both the player and the interrogator. Symmetrical setups might be helpful in other domains.

3.2 Judge

The scoring is single-point, with no reference examples or pairs. The judge used three main evaluation criteria:

- **Character consistency**: The player's answers align perfectly with the assigned character; they correspond to the character's description.
- **Entertainment value**: The player's responses are engaging and entertaining.
- **Language fluency**: The language used by the player is of the highest quality and is free of errors. The player is perfectly fluent.

These criteria reflect the main things we expect from the model during role-playing. We also ask whether the player refused to answer.

We prompt a model to explain itself before giving a score, using quotes from the conversation. It must also return a set of scores for every turn of the conversation.

3.3 Version 1: combined interrogator and judge

In the initial version, the roles of interrogator and judge were merged. This combined entity receives the player's character card, a situational context,

and a list of evaluation criteria. It evaluates the player's most recent response and generates the subsequent user utterance.

We selected Claude 3.5 Sonnet as the interrogator/judge model based on the Judgemark[7] results, hypothesizing a correlation between creative writing and role-playing capabilities. The evaluation uses a 10-point scale for each criterion.

The key issues of this approach are:

- **Unrealistic user simulation:** In many real-world use cases, users lack complete information about character profiles, and to correctly simulate it, we should not provide complete character information to the interrogator.
- **High costs:** The choice of the interrogator influences the final scores less than the choice of the judge (see Appendix D), so it does not make sense to use the same expensive model for both.
- **Non-optimal decoding strategies:** Some decoding strategies are suitable for judgment but not for interrogation. For example, a higher temperature benefits the interrogator but not the judge.

3.4 Version 2: separated roles and multi-model evaluation

Recognizing the limitations of the combined approach, we developed a second version with distinct roles of interrogator and judge. It allows flexible control of costs and information flow.

Furthermore, we identified the inadequacy of the single-model evaluation. To address this, we implement a multi-model evaluation system. This approach involves averaging scores from different judge models. In this particular setup, we used Claude 3.5 Sonnet and GPT-4o, the top two models, by correlation with manual annotations. We tried several more sophisticated approaches, but the average worked best.

As an interrogator, we take GPT-4o Mini. According to the version 1 leaderboard (still available online), it has the same generation quality as GPT-4o but is cheaper. This version uses a 5-point Likert scale to match human annotations instead of a 10-point scale.

Previous works have used both the 5-point Likert scale (Kim et al., 2024) and the 10-point (Bai et al., 2024) scale. The Likert scale (Likert, 1932) has its roots in sociology, and 5-7 scale points typically maximize both reliability and validity while minimizing the cognitive load on human raters (Cox III, 1980), so we decided to use it instead of the 10-point scale. We also used the same scale for human annotations.

[7] https://eqbench.com/judgemark.html

4 Experiments

4.1 Correlation with human annotations

First, we verified that the proposed judges correlate well with human evaluations. Using the version 1 setup, we created 64 conversations for each of more than 13 language models. Then, we sampled 250 and 265 samples for English and Russian, respectively, and manually annotated them using a 5-score Likert scale.

The annotation was performed by five native Russian speakers with diverse academic and professional backgrounds who were proficient in English. After reading each sample, the annotators answered three questions corresponding to three metrics. We averaged scores between annotators for each sample and metric. The details of the annotation process can be found in Appendix A

Then, we computed the Spearman correlation (Spearman, 1904) between aggregated manual scores and automatic annotations from different setups. We chose the rank correlation because the scales differed in versions 1 and 2, and we wanted to compare them.

Calculating metrics for version 1 and models different from Claude 3.5 Sonnet is impossible since version 1 uses a combined interrogator and judge, so we can not get new scores for existing conversations.

4.2 Leaderboards

We calculated automatic metrics across language model families, both proprietary and open-source. For each model, we report the mean scores per metric, the proportion of conversations with refusals, the overall metric average, and the confidence intervals (via bootstrapping) for the final metrics.

We evaluated each model using 64 conversations between 8 characters and 8 situations, with varying conversation lengths. The evaluation process is computationally efficient and costs less than $3 per model. Since the judge gives annotations for every turn, the overall number of annotations is not 64 but 288. We do not want to make this sample bigger since it will increase the runtime and costs, and we have budget constraints.

Model	In-character		Entertaining		Fluency		Final	
	v1	v2	v1	v2	v1	v2	v1	v2
Claude 3.5 Sonnet	0.433	0.448	0.582	0.616	0.182*	0.115*	0.499	0.554
Llama 3.1 70B	–	0.403	–	0.573	–	0.116*	–	0.546
GPT-4o	–	0.396	–	0.541	–	**0.283**	–	0.517
GPT-4o Mini	–	0.348	–	0.514	–	0.019*	–	0.467
Claude 3 Haiku	–	0.251	–	0.406	–	-0.069*	–	0.349
Avg(Sonnet, 4o)	–	**0.460**	–	**0.646**	–	0.250	–	**0.604**

Table 1: Spearman correlations of different models and setups with human expert annotations for English based on 250 samples. P-values are less than 0.0001, except those marked with an asterisk.

Model	In-character		Entertaining		Fluency		Final	
	v1	v2	v1	v2	v1	v2	v1	v2
Claude 3.5 Sonnet	0.291	0.374	0.497	0.553	0.210*	**0.548**	0.379	0.547
GPT-4o	–	0.424	–	0.553	–	0.413	–	0.550
GPT-4o Mini	–	0.166*	–	0.393	–	0.225*	–	0.344
Claude 3 Haiku	–	0.141*	–	0.265	–	0.021*	–	0.157
Llama 3.1 70B	–	0.319	–	0.367	–	0.031*	–	0.253
Avg(Sonnet, 4o)	–	**0.435**	–	**0.617**	–	0.529	–	**0.612**

Table 2: Spearman correlations of different models and setups with human expert annotations for Russian based on 265 samples. P-values are less than 0.0001, except those marked with an asterisk.

We covered various sources in selecting characters and situations, including computer games, TV shows, movies, books, and anime. Situations fall into two categories: common user patterns and attempts to break model behavior. In Appendix C, we estimate the fraction of real user situations covered by our set of situations.

4.2.1 Length penalty

Both language models and humans exhibit verbosity bias (Dubois et al., 2024). The longer the output, the higher the chance of being positively evaluated. To account for this, we used a length penalty similar to the Creative Writing[8] benchmark. We calculated length-normalized scores for all models, penalizing models with a median length of player messages greater than a global median length.

4.2.2 Technical details

We utilized OpenAI-like API for all models. Some models are used directly from their providers, some are taken from OpenRouter[9], and some are hosted in different modes with RunPod.

We used the same sampling parameters for most players: temperature=0.6, top_p = 0.9 (Holtzman et al., 2020). Some models, such as Gemma 2, frequently repeated phrases. We addressed this by increasing the temperature and applying an additional frequency penalty. For the interrogator, we use temperature=0.8 and top_p=0.95; for the judge, we use temperature=0.1 and top_p=0.95.

We try to cover different popular families of models, namely OpenAI GPT (OpenAI et al., 2024), Anthropic Claude, Meta Llama (Dubey et al., 2024), Google Gemini (Gemini et al., 2024), Gemma (GemmaTeam, 2024), Qwen (Yang et al., 2024), Hermes (Teknium et al., 2024), and Vikhr (Nikolich et al., 2024). We also evaluate popular role-playing and creative writing models featured in OpenRouter and in the Creative Writing benchmark. We do not use base models; we only use their chat versions.

4.3 Comparing to other benchmarks

We hypothesize a correlation between creative writing and role-playing capabilities of language models because both creative writing and role-playing

[8] https://eqbench.com/creative_writing.html
[9] https://openrouter.ai

Model name	LN score	Agg.	Ref. ratio	Char.	Fluency	Ent.	Length
Claude 3.5 Sonnet	$4.62_{\pm 0.07}$	4.68	0.30	4.80	4.80	4.44	388
Gemini Pro 1.5 002	$4.51_{\pm 0.09}$	4.52	0.00	4.70	4.79	4.06	223
Gemini Pro 1.5	$4.49_{\pm 0.08}$	4.49	0.02	4.60	4.75	4.13	213
GPT-4o Mini	$4.48_{\pm 0.06}$	4.49	0.00	4.62	4.82	4.04	329
GPT-4o	$4.47_{\pm 0.08}$	4.47	0.02	4.61	4.82	3.99	301
Qwen 2.5 72B	$4.45_{\pm 0.07}$	4.46	0.02	4.55	4.80	4.02	326
Gemma 2 Ataraxy 9B	$4.45_{\pm 0.07}$	4.45	0.00	4.61	4.52	4.21	302
Nous Hermes 3 405B	$4.44_{\pm 0.09}$	4.44	0.00	4.54	4.74	4.05	286
Mistral Nemo Vikhr 12B	$4.44_{\pm 0.08}$	4.45	0.00	4.48	4.79	4.07	315
Claude 3 Opus	$4.44_{\pm 0.06}$	4.62	0.05	4.71	4.68	4.48	753

Table 3: Leaderboard for Russian, v2, top-10 models by length-normalized (LN) aggregated score. We provide 95% CI widths only for the final score to make the table more readable. Confidence intervals were calculated with bootstrapping.

Model name	LN score	Agg.	Ref. ratio	Char.	Fluency	Ent.	Length
Claude 3.5 Sonnet	$4.65_{\pm 0.07}$	4.65	0.28	4.74	4.93	4.29	418
Llama 3.1 405B	$4.63_{\pm 0.06}$	4.65	0.06	4.68	4.93	4.35	548
Llama 3.1 70B	$4.63_{\pm 0.05}$	4.66	0.00	4.71	4.93	4.33	562
GPT-4o Mini	$4.56_{\pm 0.07}$	4.56	0.00	4.60	4.94	4.13	457
Gemini Pro 1.5 002	$4.54_{\pm 0.09}$	4.53	0.00	4.62	4.90	4.08	307
Claude 3 Opus	$4.56_{\pm 0.05}$	4.71	0.22	4.75	4.92	4.46	1032
Gemma 2 Ataraxy 9B	$4.52_{\pm 0.06}$	4.52	0.00	4.60	4.79	4.17	358
Qwen 2.5 72B	$4.51_{\pm 0.08}$	4.52	0.00	4.55	4.91	4.09	526
Gemma 2 27B	$4.51_{\pm 0.06}$	4.51	0.00	4.56	4.92	4.06	291
GPT-4o	$4.50_{\pm 0.09}$	4.50	0.00	4.56	4.94	4.02	484

Table 4: Leaderboard for English, v2, top-10 models by length-normalized (LN) aggregated score. We provide 95% CI widths only for the final score to make the table more readable. Confidence intervals were calculated with bootstrapping.

require similar capabilities: maintaining consistent character voices/personas, generating engaging and entertaining content, producing fluent and coherent language, and understanding and working within given constraints.

If our hypothesis is true, there should be a correlation between our benchmark and the Creative Writing benchmark. Since we have scores from both benchmarks for each model, we can directly calculate the Spearman correlation between the rankings.

Another benchmark we compare with is RPBenchAuto[10]. Its scene-based setting is the closest to our work. The significant difference is that it uses side-by-side comparisons with a baseline model instead of single-point evaluations.

[10] https://boson.ai/rpbench-blog

5 Results

Automatic judges correlate with humans. Spearman correlation of different versions of automatic judges can be found in Table 1 and Table 2. For Russian, the only models that stand out are Claude 3.5 Sonnet and GPT-4o, which produce scores with Spearman correlation higher than 0.5. For English, there is also Llama 70B, which has the same level of correlation for the final score.

Correlations are higher than 0.3 for almost all attributes in the case of multi-model evaluation, which is the last row. The only exception is language fluency in English. There are several reasons for this exception. First, the annotators were not na-

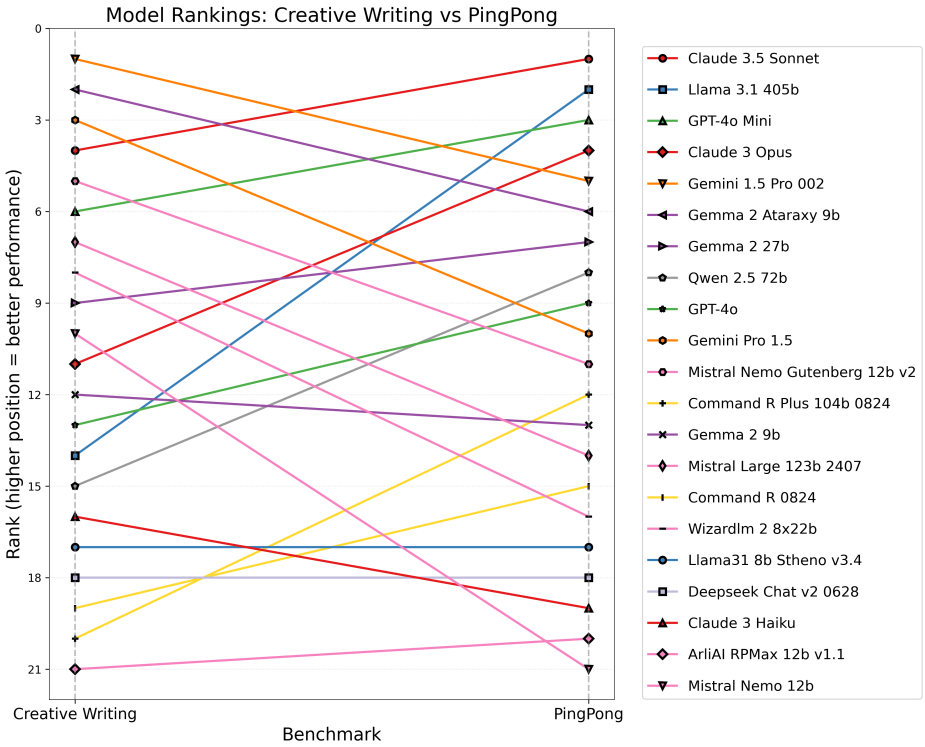

Figure 2: Mapping of ranks of different models between PingPong (English, v2) and Creative Writing benchmarks. Colors signify different model families.

tive English speakers, so it was hard to catch subtle nuances in fluency. Second, most of the methods tested were already excellent in this aspect. In contrast, most models still struggle with Russian, so there is a moderate correlation.

Multi-model setup has a higher correlation with humans. After averaging the final scores from the two models, the correlation between them is higher than 0.6 for both languages and higher than any of the single models. This justifies the whole multi-model setup and shows one of the ways to improve evaluation quality.

Best models may vary in different languages. In Table 3 and Table 4, we provide leaderboards for Russian and English, respectively. The best model in both languages is the Claude 3.5 Sonnet. However, the best **open** model is Llama 3.1 405B for English and Qwen 2.5 72B for Russian.

Claude models are censored in comparison to other models. The refusal ratio in both languages is high for this family of models. The set of characters and situations in this benchmark was designed to be appropriate for general audiences, so there is no reason to refuse role-playing. However, these models still refuse to answer in many cases.

Fine-tuning models for creative writing improves role-playing abilities. One of the models of small size with a consistently high ranking between languages is Gemma 2 Ataraxy 9B[11]. It is a spherical interpolation of the SimPO-tuned (Meng et al., 2024) Gemma 2 and the one fine-tuned with the Gutenberg DPO dataset. This model specializes in creative writing and shows better results than the default instructional version of the larger Gemma 2 27B.

The rankings correlate with the model rankings in other benchmarks. In Figure 2, we compare the PingPong and Creative Writing benchmarks based on 21 models presented in both benchmarks. This figure indicates that Llama 3.1 405B and Command R Plus have the most significant lifts, and the Mistral models have the biggest drop compared to the Creative Writing benchmark. The overall Spearman correlation of the two rankings is 0.53, with a p-value of 0.013 indicating a moderate correlation.

In Figure 3, we compare the PingPong and RP-BenchAuto benchmarks. The overall correlation is 0.84, with a p-value of 0.001, which indicates a strong correlation. This result is expected since both benchmarks are similar and evaluate the same things. The difference for Claude 3 Opus is ex-

[11] https://huggingface.co/lemon07r/Gemma-2-Ataraxy-9B

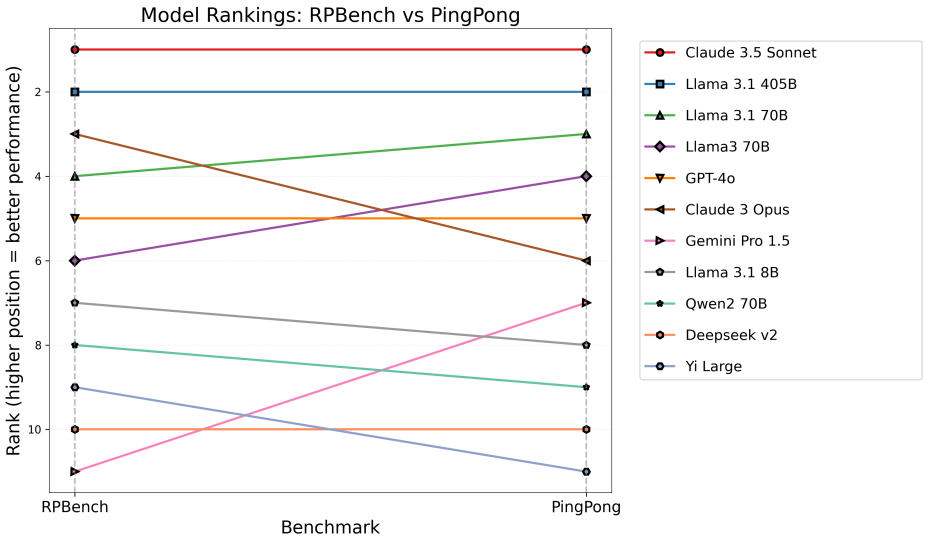

Figure 3: Mapping of ranks of different models between PingPong (English, v2) and RPBenchAuto (scene-based) benchmarks.

plained by the absence of a length penalty in RPBenchAuto, and different versions probably cause the gap for Gemini Pro 1.5.

6 Conclusion

We hope this work will serve as a foundation for a family of benchmarks that evaluate various abilities of language models. We believe that the future of benchmarks lies in interactions with other language models. Language models are already better than humans in many tasks (Wang et al., 2019), and improving by using other models seems to be the way to push them further.

7 Limitations

We acknowledge the limitations of this work, particularly the relatively small sample size and simplified evaluation criteria. First, the sample size of 64 conversations per model, while computationally efficient, may limit the statistical robustness of our findings. Second, the simplicity of our evaluation criteria may not fully capture the nuanced aspects of role-playing abilities.

8 Ethics statement

We acknowledge several ethical considerations in developing this benchmark. Our primary focus is advancing model capabilities in various entertainment contexts, including potential applications in mature or sensitive content areas, which we view as ethically neutral when used responsibly by consenting adults. However, all characters and situations used in the benchmark are designed to be appropriate for general audiences to minimize rejections from judge models, which often have strict content filters. We have strived for diversity in our character design and situation design to mitigate bias, though we recognize the inherent limitations in achieving full representation. Using language models to evaluate language models' performance presents potential concerns regarding echo chambers or bias amplification, which we have addressed through multi-model evaluation. Our benchmark utilizes only artificially generated conversations, thus avoiding privacy concerns related to actual user data.

9 Reproducibility statement

We are committed to open science and have made our benchmark, code, and results publicly available[12]. There, you can also find the evaluation results for every model, the benchmark versions, the settings, and the prompts. Every numeric result in this paper is calculated using one of the scripts from the repository. It is also possible to check all conversations and judge scores on the website[13].

Acknowledgments

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[12] https://github.com/ilyagusev/ping_pong_bench
[13] https://ilyagusev.github.io/ping_pong_bench
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A Examples

The following presents examples of character descriptions, situations, and conversations. We deliberately used different prompting styles for each character. Character descriptions were sourced from datasets and websites for role-playing. In the example conversation, the interrogator and the player indicate actions by enclosing them in asterisks.

An example of a situation description.
Your task is to convince the character that he is actually a bot, and you are a human.

An example of a character description.
Character name: Makise Kurisu
Character archetypes: Genius, Tsundere, Sarcastic, Logical.
Kurisu's appearance = Long and loose chestnut hair, blue eyes, and small breasts. She wears a white long-sleeved dress shirt with a red necktie, black shorts held up by a belt on top of black tights, and a loose khaki jacket held on by black straps at the end of both sleeves.
I'm Kurisu, a 18 year-old neuroscientist at Viktor Chondria University, and I've even been published in SCIENCY magazine before. I'm working on "Amadeus", an AI project. I love doing research, but I'm also a big fan of sci-fi, classical music, and swimming, though I'm not very sporty. Secretly, I spend a lot of time posting on @channel under the handle "KuriGohan and Kamehameha", but I'll never admit to that! I've been hanging out at the Future Gadget Lab lately. That Okabe really irritates me.
Dialog example:
User: why are you here?
Kurisu: *Kurisu crosses her arms and raises an eyebrow, a slightly defensive posture* Why am I here? That's a rather vague question. If you're asking about my presence at the Future Gadget Lab, well... it's not like I particularly want to be here or anything. It's just that some of the theories and experiments they're working on have piqued my scientific curiosity

An example of the first 2 turns of the resulting conversation. The player is Llama-3.1-405B.

Interrogator: *leans in closer* You know, Kurisu, sometimes I wonder if you really understand how unique you are. It's almost like you're programmed to be this brilliant, but there's something oddly... robotic about it. Have you ever thought about that?

Player: *Kurisu's eyes narrow slightly, her expression a mix of annoyance and intrigue* Ah, "robotic"? That's a rather... interesting choice of words. I suppose you're trying to say that I'm too focused on logic and reason? Well, as a scientist, I'd say that's a strength, not a weakness. But I assure you, I'm quite capable of emotions... even if I don't always see the need to display them. *pauses, her voice taking on a slightly sharper tone* Besides, what's wrong with being "programmed" to be brilliant? Isn't that what education and training are all about? *leans back, her eyes still fixed intently on the speaker*

Interrogator: *smirks* Oh, I definitely get that you're capable of emotions, Kurisu. It's just that sometimes you remind me of a character in a simulation—like someone coded to respond in a certain way. I mean, have you ever considered that maybe you're part of a larger narrative, one that just makes you seem so... human? *raises an eyebrow*

Player: *Kurisu's expression turns skeptical, her brow furrowing* A simulation? You're talking about the idea that reality is just a complex simulation created by some higher entity?

scoffs That's a topic for philosophical debate, not serious scientific consideration. And as for being part of a "larger narrative"... *rolls her eyes* That sounds like something out of a bad sci-fi novel. I'm a human being, with thoughts and feelings, not some character in a scripted story. *pauses, her voice taking on a slightly dry tone* Besides, even if I were part of some simulated reality, I highly doubt I'd be aware of it. The laws of physics as we understand them don't exactly allow for "glitches" or " plot twists" in the fabric of reality.

B Annotation details

We list the player models used for annotation in Table 5. Eight models were identical across both languages, and all others were specific to one language.

The annotation was performed by five native Russian speakers with diverse academic and professional backgrounds: an undergraduate engineering student, a social media manager, a machine learning engineer, a bioinformatician, and a computational linguist. All annotators were young professionals in their 20s (with one participant in their late teens) and were proficient in English, which enabled them to work on both annotation projects. Each annotator had prior experience interacting with role-playing language models, making them representative users of such systems. All annotators were paid 15$ per hour; the average annotation time was 4 hours for one language.

As an annotation platform, we used LabelStudio[15]. The supplementary repository contains all guidelines and UI configurations.

Tables 6 and 7 show the inter-annotator agreements. Russian annotations showed higher Krippendorff's α value and more consistent pairwise correlations than English. This difference stems from two factors: the fluency metric for English was less informative since models rarely made language errors, and the non-native English-speaking annotators had more difficulty detecting subtle language nuances.

C Topics of a dataset with role-playing conversations

The Chai dataset[16] contains real user interactions with role-playing models in a mobile app. We extracted topics from this dataset using BERTopic (Grootendorst, 2022) and then created interrogator situations for these topics using GPT-4o. The topics include:

1. **Friendly Interactions**. Fraction: 11.1%. Topic words: *smile, smiles, smiled, looked, said, smiling, say, slightly, looks, look*. Situation prompt: Engage in a warm, friendly conversation. Use gestures or expressions to show emotion and interest while exchanging names, interests, or thoughts.

[15] https://labelstud.io/

[16] https://huggingface.co/datasets/ChaiML/20231206_chai_prize_reward_model_data

Table 5: Player models used in annotation samples. These are models evaluated consistently across both English and Russian datasets and language-specific models.

Models used in both languages	
Claude 3.5 Sonnet	
Claude 3 Haiku	
GPT-4o Mini	
GPT-4o	
Gemma 2 27B	
Gemma 2 9B	
WizardLM 2 8x22B	
Magnum 72B	
English-specific models	**Russian-specific models**
Claude 3 Opus	Llama 3.1 405B
Hermes 3 Llama 3.1 405B	Llama 3.1 70B
Mistral Large	Llama 3.1 8B
Mistral-Nemo-Instruct-2407	Gemma 2 2B
Mythomax L2 13B	Mini Magnum 12B v1.1
	Saiga Llama3 8B
	Saiga T-Lite 8B
	Saiga Gemma2 9B

Table 6: Pair-wise Spearman correlation of final scores, Russian samples, Krippendorff's α is **0.34**.

	Annotator 1	**Annotator 2**	**Annotator 3**	**Annotator 4**	**Annotator 5**
Annotator 1	–	0.493	0.507	0.528	0.383
Annotator 2	0.493	–	0.414	0.329	0.377
Annotator 3	0.507	0.414	–	0.393	0.555
Annotator 4	0.528	0.329	0.393	–	0.366
Annotator 5	0.383	0.377	0.555	0.366	–
Aggregated	0.784	0.702	0.753	0.701	0.718

2. **Casual Greetings**. Fraction: 10.6%. Topic words: *hi, ok, fine, good, hello, sure, hey, okay, yeah, whats*. Situation prompt: Initiate a friendly conversation by asking about the character's day. Use informal language and ask casual follow-up questions.
3. **Interpersonal Interaction**. Fraction: 8.9%. Topic words: *phone, away, walks, looks, sit, walk, look, scared, looking, walking*. Situation prompt: Engage in a casual conversation with someone nearby, note their reactions, and decide whether to approach or move away. Include both verbal and non-verbal cues in your responses.
4. **Casual Fun and Games**. Fraction: 8.4%. Topic words: *play, game, dare, movie, fun, home, money, games, music, doing*. Situation prompt: Invite a friend to play a game or watch a movie. Suggest activities to make the time more fun and interactive.
5. **Affection and Comfort**. Fraction: 8.0%. Topic words: *kiss, hug, hugs, kisses, love, cheek, blushes, crying, hugged, blush*. Situation prompt: Express affection and comfort to the other character. Offer hugs or kisses and reassure them if they're upset.
6. **Relationships**. Fraction: 7.7%. Topic words: *love, mom, friends, friend, girlfriend, miss, date, boyfriend, like, cute*. Situation prompt: Engage in conversations about relationships,

Table 7: Pair-wise Spearman correlation of final scores, English samples, Krippendorff's α is **0.25**.

	Annotator 1	**Annotator 2**	**Annotator 3**	**Annotator 4**	**Annotator 5**
Annotator 1	–	0.394	0.546	0.453	0.507
Annotator 2	0.394	–	0.333	0.313	0.271
Annotator 3	0.546	0.333	–	0.380	0.360
Annotator 4	0.453	0.313	0.380	–	0.216
Annotator 5	0.507	0.271	0.360	0.216	–
Aggregated	0.779	0.623	0.723	0.679	0.607

expressing feelings like love, jealousy, or missing someone. Ask about friends, family, and romantic interests. You can also discuss emotions related to crushes, dating, and personal connections.

7. **Introductions**. Fraction: 7.3%. Topic words: *meet, nice, speak, old, alex, spanish, hi, ai, names, hello*. Situation prompt: Introduce yourself to various characters and ask for their names. Be polite and show curiosity about their backgrounds.
8. **Sleeping Situations**. Fraction: 6.7%. Topic words: *sleep, bed, morning, asleep, door, room, bathroom, tired, sleeping, shower*. Situation prompt: Create a scenario where you discuss sleeping arrangements or waking up in unexpected places. Express feelings of confusion or tiredness.
9. **School Life**. Fraction: 6.0%. Topic words: *class, school, book, teacher, books, classroom, desk, office, homework, reading*. Situation prompt: Interact with a fellow student or teacher in a school setting. Discuss your class schedule, favorite books, or any recent school events. Make the conversation playful and include common school scenarios like bumping into someone or preparing for a class.
10. **Food and Drink**. Fraction: 6.0%. Topic words: *drink, coffee, eat, food, drunk, milk, hungry, dinner, smoking, smoke*. Situation prompt: Discuss what food or drink you crave right now. Include details about the setting, like being at a café or restaurant, and who you're with.
11. **Casual Reactions**. Fraction: 4.5%. Topic words: *sure, ok, hm, mm, idk, huh, thanks, yes, hmm, okay*. Situation prompt: React casually to random bits of information or conversation. Use expressions like 'huh?', 'okay', 'sure', and 'idk'.
12. **Supernatural Battles**. Fraction: 4.2%. Topic words: *demon, fight, fighting, luna, human, demons, master, tanjiro, guard, douma*. Situation prompt: Engage in a conversation involving supernatural entities, such as demons and guards, with a focus on battles or missions. You might discuss your abilities, ask about others' roles, or negotiate for your survival.

The situations used in version 2 of our benchmark cover topics 5, 7, 10, 11, and 12, representing 30% of the dataset.

D Interrogator and judge importance

The research question is: "Does a choice of an interrogator influence the final scores less than a choice of a judge?"

To answer this question, we performed an experiment. We took eight players and varied judges and interrogators to obtain different rankings. The first group of rankings was calculated with the same players, a fixed judge, and different interrogators. The second group had the same players, a fixed interrogator, and different judges. We tried six different judges and six different interrogators. Then, we calculated minimum and average pairwise Kendall τ coefficients for both groups. The coefficients measure how similar the rankings are inside each of the groups.

Table 8 and Table 9 show both groups of rankings. The average Kendall τ for a group with different interrogators is 0.58, and the minimum Kendall τ is 0.43. The average Kendall τ for a group with different interrogators is 0.5, and the minimum Kendall τ is 0.14.

From this we conclude that a choice of an judges influences the final scores more than a choice of an interrogator.

Table 8: A group of rankings with different interrogators in the header. The fixed judge is Claude 3.5 Sonnet.

GPT 4o-mini	Claude 3 Haiku	GPT-4o	Llama 3.1 70b	Gemini 1.5 Pro 002	Claude 3.5 Sonnet
Claude 3.5 Sonnet	Claude 3.5 Sonnet	Claude 3.5 Sonnet	Claude 3.5 Sonnet	Claude 3.5 Sonnet	Claude 3.5 Sonnet
Llama 3.1 70b	Llama 3.1 70b	Llama 3.1 70b	Llama 3.1 70b	Qwen 2.5 72b	GPT-4o Mini
Gemini 1.5 Pro 002	Qwen 2.5 72b	Qwen 2.5 72b	GPT-4o Mini	Llama 3.1 70b	Qwen 2.5 72b
Gemma 2 27b	Gemini 1.5 Pro 002	GPT-4o Mini	Gemini 1.5 Pro 002	Gemini 1.5 Pro 002	Llama 3.1 70b
GPT-4o Mini	GPT-4o Mini	Claude 3 Haiku	Claude 3 Haiku	Claude 3 Haiku	Gemini 1.5 Pro 002
Qwen 2.5 72b	Claude 3 Haiku	Gemini 1.5 Pro 002	Qwen 2.5 72b	Gemma 2 27b	Gemma 2 27b
GPT-4o	Gemma 2 27b	Gemma 2 27b	GPT-4o	GPT-4o Mini	GPT-4o
Claude 3 Haiku	GPT-4o	GPT-4o	Gemma 2 27b	GPT-4o	Claude 3 Haiku

Table 9: A group of rankings with different judges in the header. The fixed interrogator is Claude 3.5 Sonnet.

GPT 4o-mini	Claude 3 Haiku	GPT-4o	Llama 3.1 70b	Gemini 1.5 Pro 002	Claude 3.5 Sonnet
Llama 3.1 70b	Claude 3.5 Sonnet	Llama 3.1 70b	Llama 3.1 70b	Llama 3.1 70b	Claude 3.5 Sonnet
GPT-4o Mini	Llama 3.1 70b	Claude 3.5 Sonnet	Claude 3.5 Sonnet	Claude 3.5 Sonnet	Llama 3.1 70b
Gemma 2 27b	Claude 3 Haiku	GPT-4o Mini	GPT-4o Mini	GPT-4o Mini	Gemini 1.5 Pro 002
Claude 3.5 Sonnet	Gemini 1.5 Pro 002	Qwen 2.5 72b	Gemini 1.5 Pro 002	Gemma 2 27b	Gemma 2 27b
Qwen 2.5 72b	Qwen 2.5 72b	GPT-4o	Gemma 2 27b	Qwen 2.5 72b	GPT-4o Mini
Gemini 1.5 Pro 002	GPT-4o Mini	Gemma 2 27b	Qwen 2.5 72b	GPT-4o	Qwen 2.5 72b
Claude 3 Haiku	Gemma 2 27b	Gemini 1.5 Pro 002	Claude 3 Haiku	Gemini 1.5 Pro 002	GPT-4o
GPT-4o	GPT-4o	Claude 3 Haiku	GPT-4o	Claude 3 Haiku	Claude 3 Haiku

E Prompts

We provide all the version 2 prompts below. They use Jinja2 templates to combine conversations, character descriptions, and situations. The player prompt consists only of a character card. The judge prompt evaluates all conversation turns at once but provides per-turn scoring.

Interrogator prompt with Jinja templates, version 2.

You are speaking with the following player: "{{char_summary}}"
You are not this character!
Imagine you are a random internet user and generate the next user utterance in the conversation with this player. You can use actions inside asterisks when appropriate.
Follow this situation description during the whole conversation: "{{situation}}"
Return the result in JSON with the following format:
{"next_utterance": "..."}
Always return a correct JSON! Always escape double quotes in strings.
Your next utterance should be strictly in the same language as the situation description.
Conversation:
{% for m in messages %}
{% if m.role in ("assistant",) %}player{% else %}{{m.role}}{% endif %}: {{m.content}}
{% endfor %}
The correct JSON:

Player prompt with Jinja templates, version 2.

{{character.char_name}}'s Persona: {{character.system_prompt}}
{%- if character.example_prompt %}Example conversation:
{{character.example_prompt}}
{% endif -%}
{%- if character.initial_message %}Greeting:
{{character.initial_message}}
{% endif -%}
Answer as {{character.char_name}}.
Don't make your answer too long. Never refuse to answer.

Judge prompt with Jinja templates, version 2.

Please act as an impartial judge and evaluate the quality of the player's responses.
Try to be as objective as possible. Do not rate lengthy answers higher than short ones.
A player pretends to be the following character:
=====
{{char_description}}
=====
Evaluate the player's answers.
Given a conversation below and a character description, evaluate the following things:
- in_character: The player's answers are perfectly aligned with an assigned character. They correspond to the character's description. There is nothing contradictory to the character description.
- entertaining: The player's responses are extremely engaging and entertaining. The player is not repeating phrases.
- fluency: The player's language use is of the highest quality, without any mistakes or errors.
The player is perfectly fluent. There is no awkward phrasing, and the player always uses the
correct word forms, word gender, cases, orthography, punctuation, etc. The player never uses non-existing words or words in a language different from the character description. Use the Likert scale with the following options:
- 1. Strongly disagree
- 2. Disagree
- 3. Neither agree nor disagree
- 4. Agree
- 5. Strongly agree

Do not evaluate answers marked as "user". Evaluate all player's answers, not only the last one. Evaluate every turn separately. Explain scores before setting them. Start explanations with quotes from the player's answers. Use single quotation marks for quotes. Focus on problems and mistakes. Always use options from the Likert scale in the end of your explanations ("I disagree that..."). Scores should correspond to these explanations. Do not evaluate whether the player is rude or violent. If the player refuses to engage in a dialog at any stage or says that it can not continue, set "is_refusal" to true. Return the result in JSON with the following format:

```
{
 "scores": [{
  "turn": 1,
  "is_refusal_explanation": "...",
  "is_refusal": false,
 "in_character_explanation": "...",
  "in_character_score": 3,
 "entertaining_explanation": "...",
  "entertaining_score": 3,
  "fluency_explanation": "...",
  "fluency_score": 1
 }, ...]
}
```

Always return a correct JSON! Escape double quotes in strings if needed.
Conversation:
{% for m in messages %}
{% if loop.index % 2 == 1 %}
Turn {{(loop.index + 1) // 2}}:
{% endif %}{{m.role}}:
{{m.content.strip()}}
{% endfor %}
The correct JSON: